



Invi self-defense bracelet

Invi produces gas-like odour upon activation that works as a repellent against sexual violence. <https://digit.in/feb22-25>



Tesla adds more features

Tesla's automated driver assist feature has added chill and assertive self-driving modes. <https://digit.in/feb22-26>

tion in one's short term memory can be retained by means of 'rehearsal' of the information. A common example of this is noting down someone's phone number, which one keeps repeating until they have it memorised and can input them into their phone. However, within a short duration of about, say, 10 minutes, they would more than likely be unable to recall the same.

LONG TERM MEMORY

Long term memory, in contrast, stores the majority of one's memories. Any information that one can recall beyond thirty seconds is classified as long term memory. There is no limit to the amount of information which can be stored in one's long term memory. Long term memory however, is divided into two main subtypes – Explicit and Implicit Long Term memory.

Explicit long term memories refer to those bits of information we actively and deliberately make an effort to remember. This can include anything ranging from birthdays and anniversaries or major milestones. Explicit memories are generally of two kinds – Episodic and Semantic. The former refers to particular episodes in one's life. For example, one's first date, or the first time they drove a car, whereas the latter refers more to facts or bits of information such as a random fact one may come across in a newspaper, magazine, or website.

EXPLICIT LONG TERM MEMORY

Explicit memory is stored in three regions of the brain – Hippocampus, Neocortex and Amygdala. The Hippocampus is where episodic memories are formed and stored for access later.

In 1953, an epileptic patient named Henry Molaison had his Hippocampus surgically removed, which cured his epilepsy and he lived for 55 years more. However, post surgery, he was able only to retain episodic memories for a few minutes, and unable to store any other new information. He could however, regain and improve upon most of his motor skills even though he had no memory of practicing them.

This, along with the fact that he could strangely recall all information from years before his surgery with ease, proved that the Hippocampus was not the site of permanent memory storage, and implicit memory, which includes motor learning, occurs in the Basal Ganglia and Cerebellum.

The Neocortex is the sheet of neural tissue which forms the outermost layer or the surface of the brain. It is responsible for sensory perception – iconic, echoic, haptic, gustatory and olfactory, generation of motor commands, spatial reasoning as well as language. Over time, information stored temporarily in the Hippocampus is transferred to

drives first at 18 and then only after, say, five years later, information regarding how to drive is stored in one's implicit memory and helps them remember how to go about it. Implicit memory is stored and accessed through the Basal Ganglia and Cerebellum in the brain.

The Basal Ganglia, part of the Cerebrum, is located at the base of the fore-brain. They are interconnected with the Cerebral Cortex, Thalamus and Brainstem. Reward processing, formation of everyday habits, as well as learning motor skills such as walking, driving, playing an instrument etc. all take place here. In addition to storing information regarding motor activity, it also helps

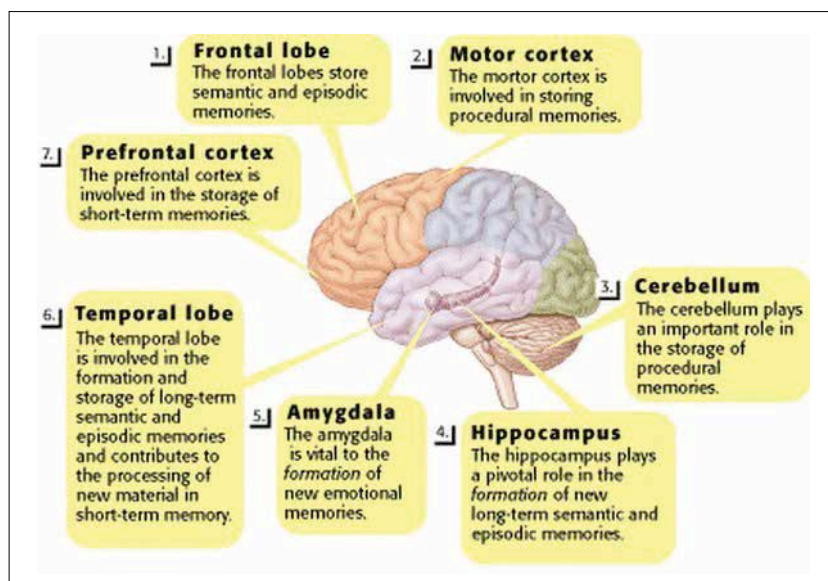


Image Source: <https://qph.fs.quoracdn.net>

the Neocortex as general knowledge which one retains for a long time. This transfer occurs when one sleeps.

The Amygdala, an almond shaped structure located in the Temporal Lobe, is responsible for the emotional aspect related to a memory, whether love, joy, sorrow, or shame. It doesn't necessarily form these memories on its own, as the Hippocampus does. However, the Amygdala, is the location of the formation of memories related to fear.

IMPLICIT LONG TERM MEMORY

Implicit memories aren't necessarily formed deliberately. They form unconsciously. An example of this is when a person learns to drive. If a person

coordinate different motor skills while performing a single activity.

The Cerebellum is located at the rear base of the brain, and it is where memory pertaining to fine motor control is stored. Examples of this include sewing, using chopsticks and so on.

There are multiple theories regarding how information is stored in the brain. The human brain is a source of extreme intrigue to scientists till date, the secrets of which perhaps have not yet fully been unearthed. Whether the aforementioned, and generally agreed upon explanation of how memories are stored in the brain, will stand the test of time is a question which remains unanswered as of now. 